

Does agroforestry modelling need a paradigm shift?

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Abstract

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Abstract

Agroforestry systems are complex. The intricate ways in which trees and annual crops, animals and other system components interact have presented major challenges for agroforestry modellers. Some progress has been made in modelling agroforestry systems (Luedeling et al. 2016), but the resulting tools have remained very difficult to use. Data requirements have been immense and simulations have usually relied on extensive field studies – or on bold assumptions. This may be acceptable wherever the goal is to predict the performance of well-established agroforestry systems, but it is not helpful for assessing agroforestry's performance in new places, or innovative agroforestry options. In such settings, collecting data is impossible, placing stifling constraints on the use of established agroforestry models for decision support. Even where datasets are available, models are often unable to produce decision-relevant information, because they do not adequately represent the inevitable uncertainty that accompanies all projections on systems that are not perfectly understood.

Agroforestry models have also struggled to address the full scope of benefits tree-based agricultural systems can produce. Obvious benefits, in particular the yield of annual crops, are usually considered, but agroforestry systems produce a wide range of somewhat more complex benefits, such as soil protection, erosion control, water cycle regulation, biodiversity conservation, microclimatic improvement, additional marketable products and intangible cultural benefits such as the aesthetic appeal of agroforested landscapes (Fig. 1). These important factors have proven difficult to model, but omitting them can lead to erroneous recommendations when models are used to support land use decisions. Agroforestry systems are likely to produce greater benefits in most respects with the possible exception of annual crop yields. Models that do not account for the full scope of system impacts risk systematically underappreciating the true benefits of adopting agroforestry practices.

Capturing the complex interactions in agroforestry systems, as well as the full scope of their impacts, has been a challenge for agroforestry modellers. In our view, the main reason for this is that researchers often have the unrealistic ambition to achieve precise understanding of all system interactions. Models are usually built through an approach that attempts to assemble all of a system's constituent parts. For each of these parts, precise relationships between system parameters and variables are defined, and the overall construct of many such pieces is then applied to simulate the performance of the overall system. This concept may work fine for relatively simple, highly predictable systems. However, it becomes problematic for complex systems like agroforestry, where outcomes of important processes cannot be predicted with precision, where variables cannot be precisely quantified or where complex outcome dimensions defy precise modelling altogether.

We propose a new paradigm for agroforestry modelling that is based on the principles of decision analysis (Luedeling and Shepherd 2016). A guiding maxim of decision analysis is that a model that is intended to

support a particular decision does not need to consider all details of the target system. Instead, it needs to account for only those aspects of the system that are decision-relevant. For each of these aspects, the state of knowledge should be captured and fully considered in generating model outputs. Possible tools to implement such computations are Monte Carlo simulations (Kuyah et al. 2019) and Bayesian Networks (Yet et al. 2016). The focus of such models is the provision of decision support, which is often achievable by collating existing information, without the need for collecting additional data beyond what is already known. Additional measurements are necessary only where model outputs based on the current state of knowledge are inconclusive. We present a framework for achieving such analyses and discuss the potential of this innovative approach to emerge as a new pragmatic paradigm for agroforestry modelling that addresses both the need for decision-relevant guidance and the prerogative of adequately reflecting the complexity and diversity of agroforestry settings.



Figure 1. Agroforestry systems such as this Acacia-tef system in Ethiopia deliver a wide range of provisioning, regulating and cultural services in addition to annual crop yields. In providing model-based support to decisions related to agroforestry adoption, such services should be considered, but common modelling approaches struggle to achieve this feat.

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